

EEE283

Digital Logic Design

Course Instructor: A K M Anindya Alam

Topic 4:

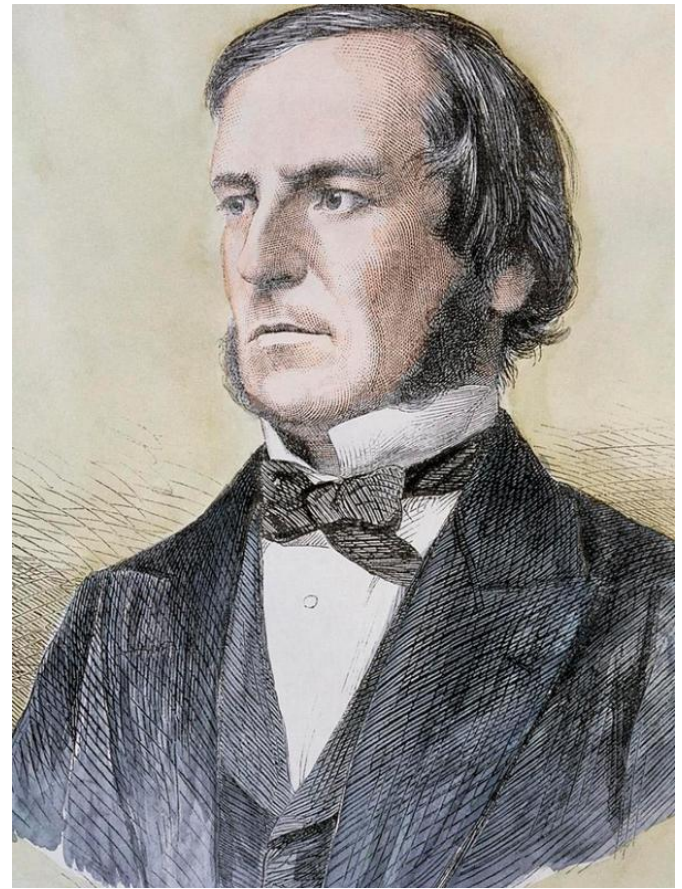
Boolean Algebra

Boolean Algebra

- ❖ Found application in engineering after almost 100 years
- ❖ Introduced by George Boole In 1849

Axioms

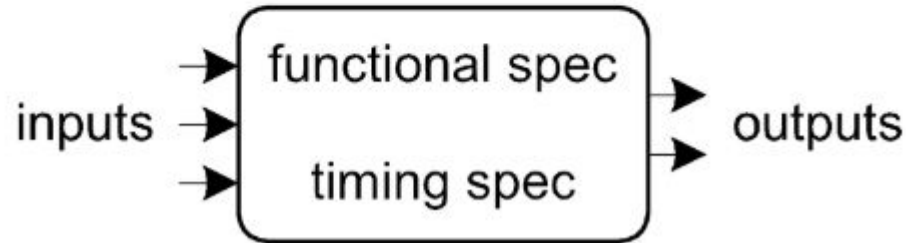
- ❖ Based on a set of rules derived from a small number of basic assumptions- Axioms
- ❖ Think of **Euclidean Geometry**



Logic Circuits

A logic circuit is composed of:

- Inputs
- Outputs
- Functional specification
- Timing specification



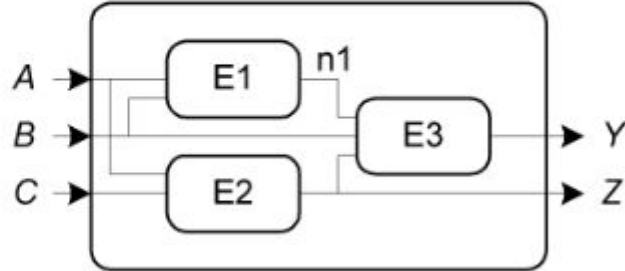
Logic Circuits

- **Nodes**

- Inputs: A , B , C
- Outputs: Y , Z
- Internal: $n1$

- **Circuit elements**

- $E1$, $E2$, $E3$
- Each a circuit



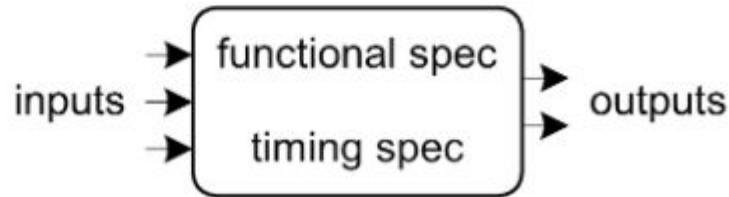
Types of Logic Circuits

- **Combinational Logic**

- Memoryless
- Outputs determined by current values of inputs

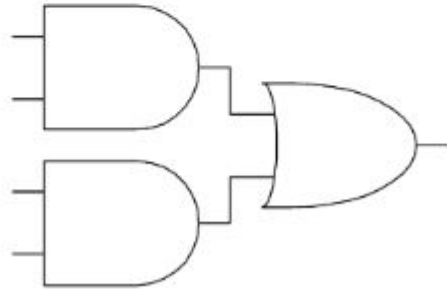
- **Sequential Logic**

- Has memory
- Outputs determined by previous and current values of inputs



Combinational Logic Circuit

- Every element is combinational
- Every node is either an input or connects to *exactly one* output
- The circuit contains no cyclic paths
- **Example:**



Each Logic Circuit is a circuit representation of a Boolean Equation which we built using Logic gates!

Fundamental Axioms of Boolean Algebra

AND/Logical Multiplication

$$0.0 = 0$$

$$0.1 = 0$$

$$1.0 = 0$$

$$1.1 = 0$$

$$A.A = A$$

$$A.0 = 0$$

$$A.1 = A$$

OR/Logical Addition

$$0+0 = 0$$

$$0+1 = 1$$

$$1+0 = 1$$

$$1+1 = 1$$

$$A+A = A$$

$$A+0 = A$$

$$A+1 = 1$$

NOT

$$\overline{0} = 1$$

$$\overline{1} = 0$$

$$\overline{\overline{A}} = A$$

Boolean Theorems for One Variable

Number	Theorem	Dual	Name
T1	$B \cdot 1 = B$	$B + 0 = B$	Identity
T2	$B \cdot 0 = 0$	$B + 1 = 1$	Null Element
T3	$B \cdot B = B$	$B + B = B$	Idempotency
T4	$\overline{\overline{B}} = B$		Involution
T5	$B \cdot \overline{B} = 0$	$B + \overline{B} = 1$	Complements

Dual: Replace: $\bullet$ with $+$
0 with 1

Boolean Theorems for Multiple Variables

#	Theorem	Dual	Name
T6	$B \cdot C = C \cdot B$	$B + C = C + B$	Commutativity
T7	$(B \cdot C) \cdot D = B \cdot (C \cdot D)$	$(B + C) + D = B + (C + D)$	Associativity
T8	$B \cdot (C + D) = (B \cdot C) + (B \cdot D)$	$B + (C \cdot D) = (B + C) (B + D)$	Distributivity
T9	$B \cdot (B + C) = B$	$B + (B \cdot C) = B$	Covering
T10	$(B \cdot C) + (B \cdot C) \bar{B} = B$	$(B + C) \cdot (B + \bar{C}) = B$	Combining
T11	$(B \cdot C) + (\bar{B} \cdot D) + (C \cdot D) = (B \cdot C) + (B \cdot D)$	$(B + C) \cdot (\bar{B} + D) \cdot (C + D) = (B + C) \cdot (B + D)$	Consensus

Dual: Replace: $\cdot$ with $+$
 0 with 1

Some Definitions

- **Complement:** variable with a bar over it
 $\bar{A}, \bar{B}, \bar{C}$
- **Literal:** variable or its complement
 $A, \bar{A}, B, \bar{B}, C, \bar{C}$
- **Implicant:** product of literals
 ABC, AC, BC
- **Minterm:** product that includes all input variables
 $ABC, \bar{A}\bar{B}\bar{C}, ABC$
- **Maxterm:** sum that includes all input variables
 $(A+\bar{B}+C), (\bar{A}+B+\bar{C}), (\bar{A}+\bar{B}+C)$

Simplifying a Boolean Equation

Reducing an equation to the **fewest number of implicants**, where each implicant has the **fewest literals**

Recall:

– Implicant: product of literals

$ABC, \bar{A}C, \bar{B}C$

– Literal: variable or its complement

$A, \bar{A}, B, \bar{B}, C, \bar{C}$

*Simplifying the equation is also called **minimizing** the equation*

Simplifying a Boolean Equation

Example 1:

$$Y = AB + A\bar{B}$$

$$Y = A \quad \text{T10: Combining}$$

or

$$= A(B + \bar{B}) \quad \text{T8: Distributivity}$$

$$= A(1) \quad \text{T5': Complements}$$

$$= A \quad \text{T1: Identity}$$

Simplifying a Boolean Equation

Example 2:

$$Y = A(AB + ABC)$$

$$= A(AB(1 + C)) \quad \text{T8: Distributivity}$$

$$= A(AB(1)) \quad \text{T2': Null Element}$$

$$= A(AB) \quad \text{T1: Identity}$$

$$= (AA)B \quad \text{T7: Associativity}$$

$$= AB \quad \text{T3: Idempotency}$$

Simplifying a Boolean Equation

Example 3:

$$Y = A'BC + A'$$

Recall: $A' = A$ —

$$= A' \quad \text{T9' Covering: } X + XY = X$$

or

$$= A'(BC + 1) \quad \text{T8: Distributivity}$$

$$= A'(1) \quad \text{T2': Null Element}$$

$$= A' \quad \text{T1: Identity}$$

Simplifying a Boolean Equation

Example 4:

$$Y = AB'C + ABC + A'BC$$

$$= AB'C + \mathbf{ABC} + \mathbf{ABC} + A'BC \quad \text{T3': Idempotency}$$

$$= (AB'C + ABC) + (ABC + A'BC) \quad \text{T7': Associativity}$$

$$= AC + BC \quad \text{T10: Combining}$$

Simplifying a Boolean Equation

Example 5:

$$Y = AB + BC + B'D' + AC'D'$$

Method 1:

$$\begin{aligned} Y &= AB + BC + B'D' + (ABC'D' + AB'C'D') && \text{T10: Combining} \\ &= (AB + ABC'D') + BC + (B'D' + AB'C'D') && \text{T6: Commutativity} \\ &\quad \text{T7: Associativity} \\ &= AB + BC + B'D' && \text{T9: Covering} \end{aligned}$$

Method 2:

$$\begin{aligned} Y &= AB + BC + B'D' + AC'D' + AD' && \text{T11: Consensus} \\ &= AB + BC + B'D' + AD' && \text{T9: Covering} \\ &= AB + BC + B'D' && \text{T11: Consensus} \end{aligned}$$

Simplifying a Boolean Equation

Example 6:

$$Y = (A + BC)(A + DE)$$

Apply T8' first when possible: $W + XZ = (W + X)(W + Z)$

Make: $X = BC$, $Z = DE$ and rewrite equation

$$Y = (A + X)(A + Z) \quad \text{substitution (X=BC, Z=DE)}$$

$$= A + XZ \quad \text{T8': Distributivity}$$

$$= A + BCDE \quad \text{substitution}$$

or

$$Y = AA + ADE + ABC + BCDE \quad \text{T8: Distributivity}$$

$$= A + ADE + ABC + BCDE \quad \text{T3: Idempotency}$$

$$= \mathbf{A + ADE} + ABC + BCDE$$

$$= \mathbf{A} + ABC + BCDE \quad \text{T9': Covering}$$

$$= A + BCDE \quad \text{T9': Covering}$$

Proving the Simplification Theorem

“Simplification” theorem
 $PA + \bar{A} = P + \bar{A}$

Method 1:

$$\begin{aligned} PA + \bar{A} &= PA + (\bar{A} + \bar{A}P) && \text{T9' Covering} \\ &= PA + P\bar{A} + \bar{A} && \text{T6 Commutativity} \\ &= P(A + \bar{A}) + \bar{A} && \text{T8 Distributivity} \\ &= P(1) + \bar{A} && \text{T5' Complements} \end{aligned}$$

Method 2:

$$\begin{aligned} PA + \bar{A} &= (\bar{A} + A)(\bar{A} + P) && \text{T8' Distributivity} \\ &= 1(\bar{A} + P) && \text{T5' Complements} \\ &= \bar{A} + P && \text{T1 Identity} \end{aligned}$$

DeMorgan's Theorem

#	Theorem	Dual	Name
T12	$\overline{B_0 \cdot B_1 \cdot B_2 \dots} = \overline{B_0} + \overline{B_1} + \overline{B_2} \dots$	$\overline{B_0 + B_1 + B_2 \dots} = \overline{B_0} \cdot \overline{B_1} \cdot \overline{B_2} \dots$	DeMorgan's Theorem

The **complement** of the **product**
is the
sum of the **complements**.

Dual: The **complement** of the **sum**
is the
product of the **complements**.

**The most important
theorem in Boolean
algebra probably!**

Application of DeMorgan's Theorem

Example 1

$$\begin{aligned} Y &= \overline{(A+BD)}\overline{C} \\ &= \overline{(A+BD)} + \overline{\overline{C}} \\ &= (\overline{A} \cdot \overline{BD}) + C \\ &= (\overline{A} \cdot (BD)) + C \\ &= \overline{A}BD + C \end{aligned}$$

Example 2

$$\begin{aligned} Y &= \overline{(\overline{ACE+D})} + B \\ &= (\overline{ACE+D}) \cdot \overline{B} \\ &= (\overline{ACE} \cdot \overline{D}) \cdot \overline{B} \\ &= ((\overline{AC} + \overline{E}) \cdot D) \cdot \overline{B} \\ &= ((AC + \overline{E}) \cdot D) \cdot \overline{B} \\ &= (ACD + D\overline{E}) \cdot \overline{B} \\ &= \overline{A}BCD + \overline{B}D\overline{E} \end{aligned}$$

Sum of Product (SOP)

- All Boolean equations can be written in SOP form
- Each row has a **minterm**
- A minterm is a **product** (AND) of literals
- Each minterm is **TRUE** for that row (and only that row)
- Form function by **ORing minterms** where output is **1**
- Thus, a **sum** (OR) of **products** (AND terms)

A	B	Y	minterm	minterm name
0	0	0	$\overline{A} \overline{B}$	m_0
0	1	1	$\overline{A} B$	m_1
1	0	0	$A \overline{B}$	m_2
1	1	1	$A B$	m_3

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$$Y = F(A, B) = \overline{A}B + AB = \Sigma(1, 3)$$

Product of Sum (POS)

- All Boolean equations can be written in POS form
- Each row has a **maxterm**
- A maxterm is a **sum** (OR) of literals
- Each maxterm is **FALSE** for that row (and only that row)
- Form function by **ANDing maxterms** where output is 0
- Thus, a **product** (AND) of **sums** (OR terms)

A	B	Y	maxterm	maxterm name
0	0	0	$A + B$	M_0
0	1	1	$A + \overline{B}$	M_1
1	0	0	$\overline{A} + B$	M_2
1	1	1	$\overline{A} + \overline{B}$	M_3

$$Y = F(A, B) = (A + B) \bullet (A' + B) = \Pi(0, 2)$$

Analyzing Boolean Eqn using SOP/POS

- You are going to the cafeteria for lunch
 - You won't eat lunch ($E=0$)
 - If it's not open ($O=0$) or
 - If they only serve corndogs ($C=1$)
- Write a truth table for determining if you will eat lunch (E).

O	C	E
0	0	0
0	1	0
1	0	1
1	1	0

SOP – sum-of-products

O	C	E	minterm
0	0		$\overline{O} \overline{C}$
0	1		$\overline{O} C$
1	0		$O \overline{C}$
1	1		$O C$

$$E = OC$$

$$= \Sigma(2)$$

POS – product-of-sums

O	C	E	maxterm
0	0		$O + C$
0	1		$O + \overline{C}$
1	0		$\overline{O} + C$
1	1		$\overline{O} + \overline{C}$

$$E = (O + C)(O + \overline{C})(\overline{O} + \overline{C})$$

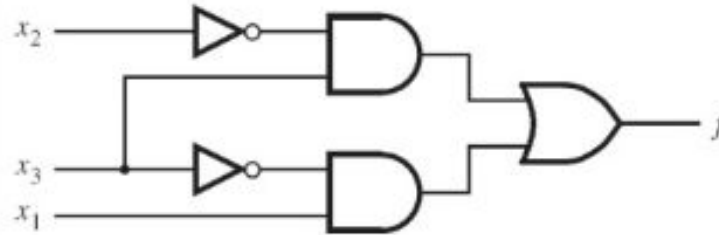
$$= \Pi(0, 1, 3)$$

Logic Circuit Synthesis from Truth Table

$$f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2x_3 + x_1\bar{x}_2\bar{x}_3 + x_1\bar{x}_2x_3 + x_1x_2\bar{x}_3$$

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

$$\begin{aligned} f &= (\bar{x}_1 + x_1)\bar{x}_2x_3 + x_1(\bar{x}_2 + x_2)\bar{x}_3 \\ &= 1 \cdot \bar{x}_2x_3 + x_1 \cdot 1 \cdot \bar{x}_3 \\ &= \bar{x}_2x_3 + x_1\bar{x}_3 \end{aligned}$$

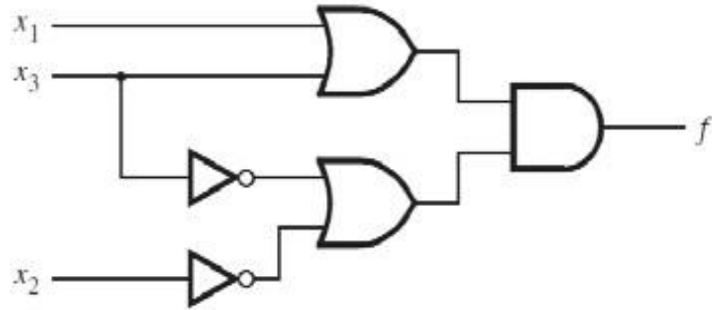


$$f(x_1, x_2, x_3) = \sum(m_1, m_4, m_5, m_6) \quad f(x_1, x_2, x_3) = \sum m(1, 4, 5, 6)$$

The $\sum$ sign denotes the logical sum operation.

Logic Circuit Synthesis from Truth Table

$$f = (x_1 + x_3)(\bar{x}_2 + \bar{x}_3)$$



Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

$$= (x_1 + x_2 + x_3)(x_1 + \bar{x}_2 + x_3)(x_1 + \bar{x}_2 + \bar{x}_3)(\bar{x}_1 + \bar{x}_2 + \bar{x}_3)$$

$$f = (x_1 + x_3)(\bar{x}_2 + \bar{x}_3)$$

Minimization of Cost of Logic Circuit

The process of finding a minimum-cost circuit involves the following steps:

- ❖ Generate all prime implicants for the given function f .
- ❖ Find the set of essential prime implicants.
- ❖ If the set of essential prime implicants covers all valuations for which $f = 1$, then this set is the desired cover of f . Otherwise, determine the nonessential prime implicants that should be added to form a complete minimum-cost cover.

The choice of nonessential prime implicants to be included in the cover is governed by the cost considerations. This choice is often not obvious.

Karnaugh Map (K-Map)

- Karnaugh map is an array of **cells** in which each cell represents a binary value of the input variables.
- The cells are arranged in a way so that simplification of a given expression is simply a matter of properly grouping the cells.
- Similar to a truth table because it presents all of the possible values of input variables and the resulting output for each value.

Karnaugh Map (K-Map)

- Boolean expressions can be minimized by combining terms
- K-maps minimize equations graphically

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Y \ C \ AB		00	01	11	10
		0	1	0	0
C	0	1	0	0	0
	1	1	0	0	0

Y C \ AB		00	01	11	10
0		$\bar{A}\bar{B}\bar{C}$	$\bar{A}B\bar{C}$	$AB\bar{C}$	$A\bar{B}\bar{C}$
	1	$\bar{A}\bar{B}C$	$\bar{A}BC$	ABC	$A\bar{B}C$

Rules of forming K-Map

Rule 1: Always fill the K-map correctly using Gray code ordering. Incorrect placement leads to incorrect minimization.

Rule 2: Groups must contain numbers of cells that are powers of two only (1, 2, 4, 8, 16...).

Rule 3: Always form the largest possible groups first to minimize literals.

Rule 4: Groups must be rectangular. Diagonal or irregular shapes are not allowed.

Rule 5: Wrap-around grouping across edges is mandatory for full optimization.

Rule 6: Overlapping groups are allowed and often necessary to reduce logic.

Rule 7: Every logic 1 must be covered by at least one group.

Rules of forming K-Map

Rule 8: Essential prime implicants must always be included.

Rule 9: Use don't-care conditions only when they help form larger groups.

Rule 10: Group cells to eliminate the maximum number of variables.

Rule 11: Write Boolean terms using only variables that remain constant in a group.

Rule 12: Avoid redundant groups that do not contribute to covering unique 1s.

Rule 13: Choose the expression with the fewest literals when multiple solutions exist.

Rule 14: Choose SOP when 1s are fewer; choose POS when 0s are fewer.

Rule 15: Perform a final sanity check to ensure all optimization rules are satisfied.

Karnaugh Map (K-Map)

- Circle 1's in adjacent squares
- In Boolean expression, include only literals whose true **and** complement form are **not** in the circle

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

$F(A,B,C) = A'B'$

Y C	AB			
	00	01	11	10
0	1	0	0	0
1	1	0	0	0

K-Map for 4 Variables

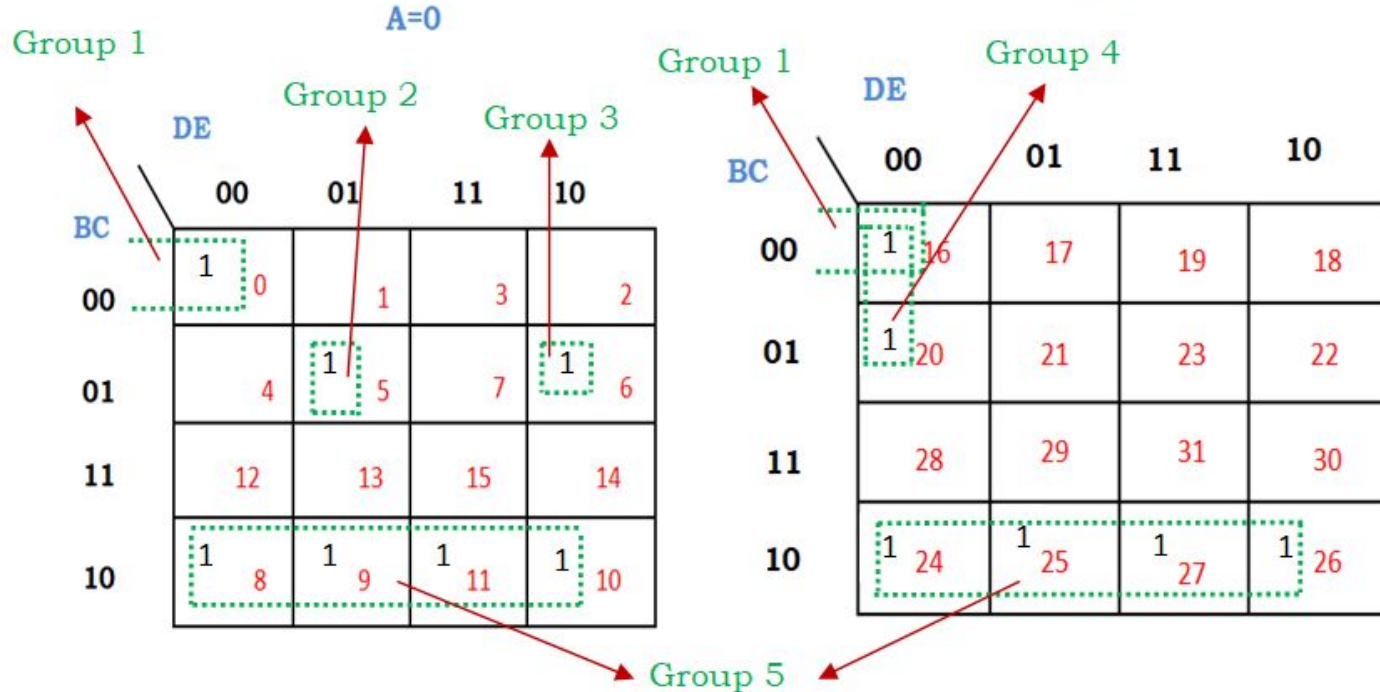
A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Y CD \ AB		AB			
		00	01	11	10
00		1	0	0	1
01		0	1	0	1
11		1	1	0	0
10		1	1	0	1

$$Y = \bar{A}\bar{C} + \bar{A}BD + A\bar{B}\bar{C} + \bar{B}\bar{D}$$

K-Map for 5 Variables

$$F(A,B,C,D,E) = \sum m(0,5,6,8,9,10,11,16,20,24,25,26,27)$$



$$F(A,B,C,D,E) = B'C'D'E' + A'B'CD'E + A'B'CDE' + AB'D'E' + BC'$$

K-Map with Don't Cares

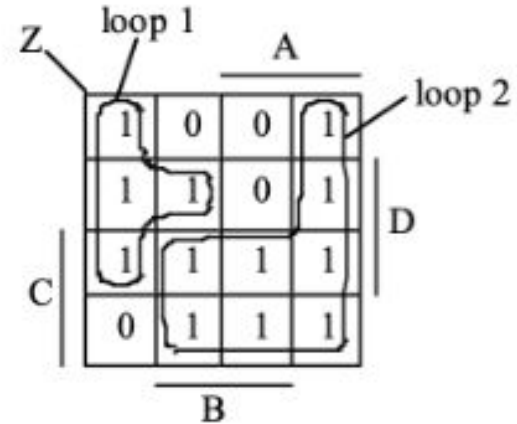
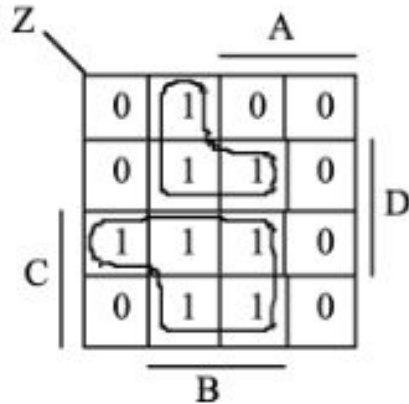
K-Maps with Don't Cares

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	X
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Y CD \ AB		AB			
		00	01	11	10
00	1	0	X	1	
01	0	X	X	1	
11	1	1	X	X	
10	1	1	X	X	

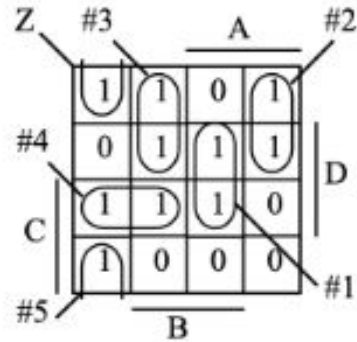
$$Y = A + \bar{B}\bar{D} + C$$

Invalid K-Map Groupings



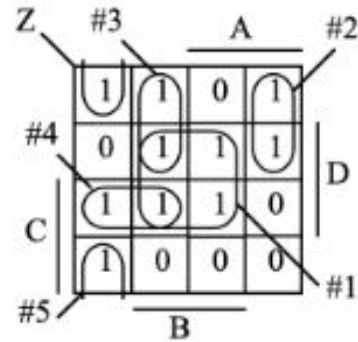
Minimal Groupings

Maximum coverage and avoid overlaps for minimal expression



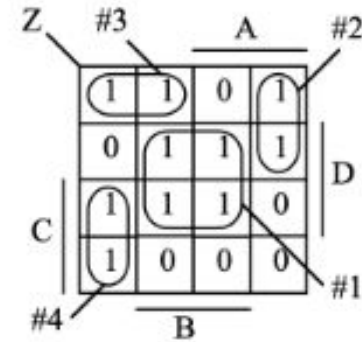
a. Non-minimal Product Terms

$$a. Z = A \cdot B \cdot D + A \cdot B' \cdot C' + A' \cdot B \cdot C' + A' \cdot C \cdot D + A' \cdot B' \cdot D'$$



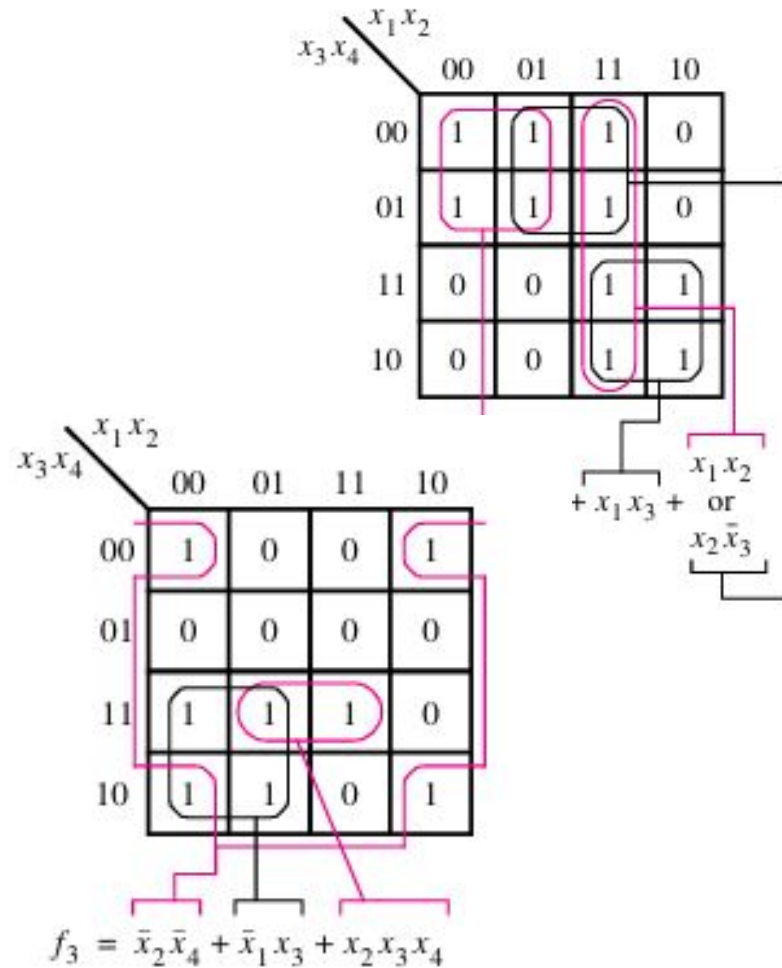
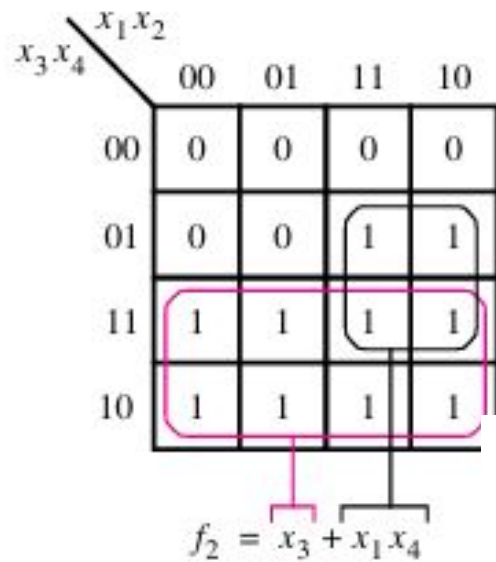
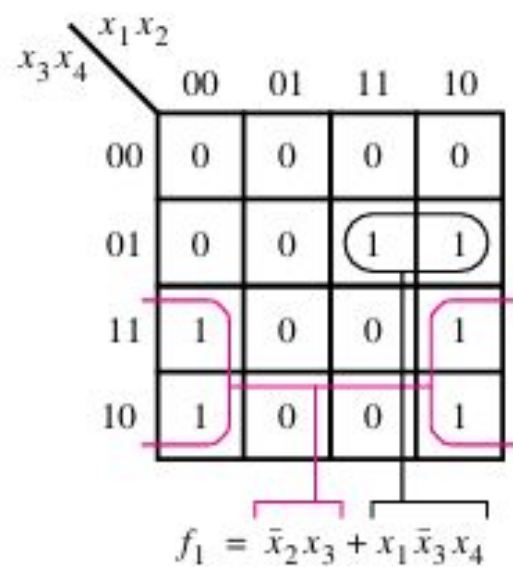
b. Non-minimal Selection of Prime Implicants

$$b. Z = B \cdot D + A \cdot B' \cdot C' + A' \cdot B \cdot C' + A' \cdot C \cdot D + A' \cdot B' \cdot D'$$



c. Minimal Grouping

$$c. Z = B \cdot D + A \cdot B' \cdot C' + A' \cdot C' \cdot D' + A' \cdot B' \cdot C$$



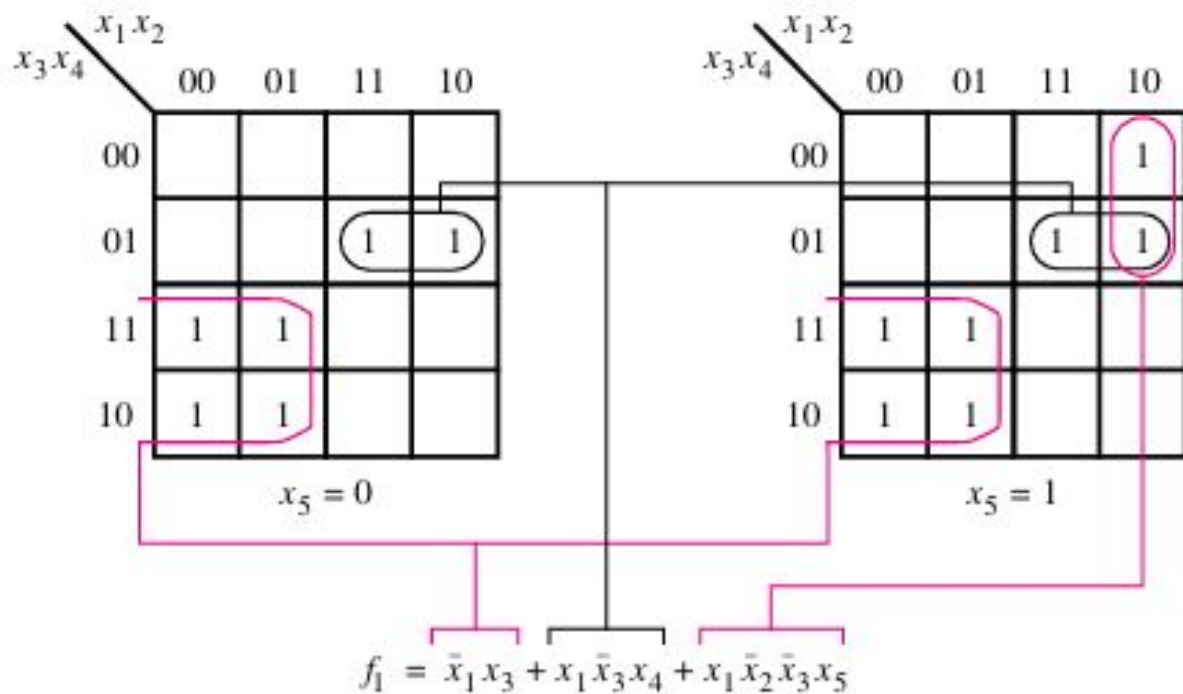


Figure 2.55 A five-variable Karnaugh map.